

RAJAT GOEL (Sr. Data Analyst | Data Scientist | AI-Expert)

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Professional Summary

Results-driven Healthcare Data Scientist with 7+ years of experience transforming complex clinical and administrative datasets into actionable insights that drive quality improvement, regulatory compliance, and operational efficiency across acute care and Medicaid settings. Currently leading nursing analytics initiatives at St. Peter's Health Partners, developing ML models that optimize care delivery while ensuring fairness and transparency in clinical decision-making. Proven expertise in Python, SQL, Azure, Snowflake, Tableau, with track record of achieving 50% efficiency gains through AI automation and 89% accuracy in predictive healthcare models. Engineered Microsoft Copilot agents for Medicaid compliance audits, reducing manpower 50% while maintaining 100% HIPAA adherence. Co-founder of ToraTax, an AI-integrated tax platform serving 3,000+ monthly visitors with 95% accuracy, demonstrating production-scale expertise in generative AI, RAG implementations, and agentic workflows.

Technical Skills

- **Data Tools:** SQL, Tableau, Power BI, Azure Data Factory, Redcap, SAS, Python (Pandas, NumPy), R, ETL tool, Claude Code, Github, Copilot
- **Healthcare Systems:** EHR/EMR (Epic, Cerner), HL7, CMS regulations, HIPAA compliance
- **Cloud Platforms:** Microsoft Azure, AWS (S3, Redshift)
- **Analytical Methods:** Predictive modeling, logistic regression, time-series analysis, data visualization
- **Consulting Skills:** Stakeholder communication, requirements gathering, client presentation

Work History

Sr. Data Analyst - Nursing Analytics

St. Peter's Health Partners | Albany, NY | Oct 2025 - Present (4 months) | Hybrid

- Built predictive ML models analyzing 50K+ nursing records identifying high-risk readmission patients, enabling targeted interventions that reduced 30-day readmission rates by 8% across 3 hospital units.
- Developed real-time nursing quality dashboard tracking falls, infections, and adverse events, enabling charge nurses to implement preventive protocols that decreased patient incidents by 12%.
- Collaborated with nursing leadership to operationalize predictive insights into clinical protocols, training 200+ nurses on data-driven risk stratification reducing escalation response time by 25%.

Data Scientist, IPRO, Albany, NY (January 2023 - Present)

- Led development of Microsoft Copilot AI personalized agents with NLP prompts to automate multi-state Medicaid compliance audits, achieving 50% manpower reduction while maintaining 100% HIPAA adherence.
- Built logistic regression models analyzing 2M+ Ohio Medicaid claims, enabling 89% accurate 30-day readmission predictions through Power BI visualizations used by 150+ care teams.
- Engineered 6 ETL pipelines via Matillion and Azure Data Factory, unifying EHR/claims data from 8 clinical domains and reducing processing time by 30% for real-time consulting reports.
- Designed ATR audit framework using Python and SAS, achieving 50% faster cycle times through automated QA checks aligned with CMS reporting standards.
- Consolidated 15+ dashboards into unified Tableau analytics hub for Pennsylvania Medicaid, enabling 40% faster stakeholder decisions through collaborative requirements gathering with clinical/operations teams.

Data Scientist Intern, AutoNation, WhitePlains, NY (July 2022 - December 2022)

- Migrated MS SQL data to AWS storage using Airflow, enabling the preparation of reports in Snowflake.
- Developed and checked ELT, row-by-row, and count tests in Query Surge to ensure the smooth running of the migration process, reducing query runtime by 40%.
- Built visually appealing and informative dashboards for clients and internal stakeholders using Tableau.

Data Analyst, Maa Ambey Traders, India (May 2016 - June 2022)

- Scraped, Analyzed, and visualized large volumes of historical food grain data using SQL, and Python to generate reports that helped the team make business decisions, resulting in a 15% increase in revenue.
- Forecasted the future demand for food grains such as wheat and pulses based on previous market supply and demand, using correlations and hypothesis testing in R on time-series data. This enabled the company to better manage its food stock and make more informed decisions.
- Developed actionable insights for the company using Excel Pivot Table by analyzing large volumes of data, which helped the sales team to make informed decisions and increase revenue by 20% per year.

EDUCATION

- PhD in Artificial Intelligence, University of Cumberlands, Dallas, Texas
- Master of Science in Data Science, Clarkson University, Potsdam, NY (2021)
- Bachelor of Commerce, Dr. Ram Manohar Lohia Awadh University, Balrampur, India (2015)

RESEARCH PROJECT, CLARKSON UNIVERSITY

Identifying Images using Neural Network algorithm (June 2022 - July 2022)

- Developed a custom CNN model using Python to detect 5 different emotions from human faces. Achieved more than 50% accuracy in detecting the correct facial expression by categorize each face based on the emotion shown in the facial expression into one of five categories (0=Angry, 1=Disgust, 2=Fear, 3=Happy, 4=Sad, 5=Surprise).

Factors affecting Coronavirus Disease (January 2022 - May 2022)

- Conducted healthcare data statistical analysis in R on age, poverty, and population, using Hypothesis testing, T-values, P- values, etc. Found a strong correlation between PM 2.5 and distance from the epicenter for infection and death. The strongest age factor for fatality is over 55, but not strong enough (R-squared=0.26). Extending the study to all the waves to see if the same variables have a strong association or if some other variable has a stronger impact.